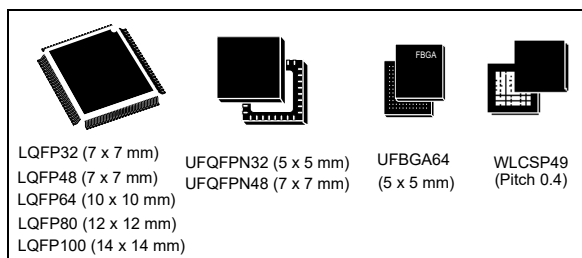


Arm[®] Cortex[®]-M4 32-bit MCU+FPU, 170 MHz / 213 DMIPS,
up to 128 KB Flash, 32 KB SRAM, rich analog, math accelerator

Datasheet - production data

Features

- Core: Arm[®] 32-bit Cortex[®]-M4 CPU with FPU, Adaptive real-time accelerator (ART Accelerator) allowing 0-wait-state execution from Flash memory, frequency up to 170 MHz with 213 DMIPS, MPU, DSP instructions
- Operating conditions:
 - V_{DD} , V_{DDA} voltage range: 1.71 V to 3.6 V
- Mathematical hardware accelerators
 - CORDIC for trigonometric functions acceleration
 - FMAC: filter mathematical accelerator
- Memories
 - 128 Kbytes of Flash memory with ECC support, proprietary code readout protection (PCROP), securable memory area, 1 Kbyte OTP
 - 22 Kbytes of SRAM, with hardware parity check implemented on the first 16 Kbytes
 - Routine booster: 10 Kbytes of SRAM on instruction and data bus, with hardware parity check (CCM SRAM)
- Reset and supply management
 - Power-on/power-down reset (POR/PDR/BOR)
 - Programmable voltage detector (PVD)
 - Low-power modes: sleep, stop, standby and shutdown
 - V_{BAT} supply for RTC and backup registers
- Clock management
 - 4 to 48 MHz crystal oscillator
 - 32 kHz oscillator with calibration
 - Internal 16 MHz RC with PLL option ($\pm 1\%$)
 - Internal 32 kHz RC oscillator ($\pm 5\%$)
- Up to 86 fast I/Os
 - All mappable on external interrupt vectors
 - Several I/Os with 5 V tolerant capability



- Interconnect matrix
- 12-channel DMA controller
- 2 x ADCs 0.25 μ s (up to 23 channels). Resolution up to 16-bit with hardware oversampling, 0 to 3.6 V conversion range
- 4 x 12-bit DAC channels
 - 2 x buffered external channels 1 MSPS
 - 2 x unbuffered internal channels 15 MSPS
- 4 x ultra-fast rail-to-rail analog comparators
- 3 x operational amplifiers that can be used in PGA mode, all terminals accessible
- Internal voltage reference buffer (VREFBUF) supporting three output voltages (2.048 V, 2.5 V, 2.95 V)
- 14 timers:
 - 1 x 32-bit timer and 2 x 16-bit timers with up to four IC/OC/PWM or pulse counter and quadrature (incremental) encoder input
 - 2 x 16-bit 8-channel advanced motor control timers, with up to 8 x PWM channels, dead time generation and emergency stop
 - 1 x 16-bit timer with 2 x IC/OCs, one OCN/PWM, dead time generation and emergency stop
 - 2 x 16-bit timers with IC/OC/OCN/PWM, dead time generation and emergency stop
 - 2 x watchdog timers (independent, window)
 - 1 x SysTick timer: 24-bit downcounter
 - 2 x 16-bit basic timers
 - 1 x low-power timer

- Calendar RTC with alarm, periodic wakeup from stop/standby
- Communication interfaces
 - 1 x FDCAN controller supporting flexible data rate
 - 3 x I²C Fast mode plus (1 Mbit/s) with 20 mA current sink, SMBus/PMBus, wakeup from stop
 - 4 x USART/UARTs (ISO 7816 interface, LIN, IrDA, modem control)
 - 1 x LPUART
 - 3 x SPIs, 4 to 16 programmable bit frames, 2 x with multiplexed half duplex I²S interface
 - 1 x SAI (serial audio interface)
 - USB 2.0 full-speed interface with LPM and BCD support
 - IRTIM (infrared interface)
 - USB Type-C™ /USB power delivery controller (UCPD)
- True random number generator (RNG)
- CRC calculation unit, 96-bit unique ID
- Development support: serial wire debug (SWD), JTAG, Embedded Trace Macrocell™

Table 1. Device summary

Reference	Part number
STM32G431x6	STM32G431C6, STM32G431K6, STM32G431R6, STM32G431V6, STM32G431M6
STM32G431x8	STM32G431C8, STM32G431K8, STM32G431R8, STM32G431V8, STM32G431M8
STM32G431xB	STM32G431CB, STM32G431KB, STM32G431RB, STM32G431VB, STM32G431MB

Contents

1	Introduction	12
2	Description	13
3	Functional overview	17
3.1	Arm® Cortex®-M4 core with FPU	17
3.2	Adaptive real-time memory accelerator (ART accelerator)	17
3.3	Memory protection unit	17
3.4	Embedded Flash memory	17
3.5	Embedded SRAM	18
3.6	Multi-AHB bus matrix	19
3.7	Boot modes	19
3.8	CORDIC	20
3.9	Filter mathematical accelerator (FMAC)	20
3.10	Cyclic redundancy check calculation unit (CRC)	21
3.11	Power supply management	21
3.11.1	Power supply schemes	21
3.11.2	Power supply supervisor	22
3.11.3	Voltage regulator	22
3.11.4	Low-power modes	23
3.11.5	Reset mode	23
3.11.6	VBAT operation	23
3.12	Interconnect matrix	25
3.13	Clocks and startup	26
3.14	General-purpose inputs/outputs (GPIOs)	27
3.15	Direct memory access controller (DMA)	27
3.16	DMA request router (DMAMux)	28
3.17	Interrupts and events	28
3.17.1	Nested vectored interrupt controller (NVIC)	28
3.17.2	Extended interrupt/event controller (EXTI)	28
3.18	Analog-to-digital converter (ADC)	29
3.18.1	Temperature sensor	29

3.18.2	Internal voltage reference (VREFINT)	30
3.18.3	VBAT battery voltage monitoring	30
3.19	Digital to analog converter (DAC)	30
3.20	Voltage reference buffer (VREFBUF)	31
3.21	Comparators (COMP)	32
3.22	Operational amplifier (OPAMP)	32
3.23	Random number generator (RNG)	32
3.24	Timers and watchdogs	32
3.24.1	Advanced motor control timer (TIM1, TIM8)	33
3.24.2	General-purpose timers (TIM2, TIM3, TIM4, TIM15, TIM16, TIM17)	34
3.24.3	Basic timers (TIM6 and TIM7)	34
3.24.4	Low-power timer (LPTIM1)	35
3.24.5	Independent watchdog (IWDG)	35
3.24.6	System window watchdog (WWDG)	35
3.24.7	SysTick timer	35
3.25	Real-time clock (RTC) and backup registers	36
3.26	Tamper and backup registers (TAMP)	36
3.27	Infrared transmitter	37
3.28	Inter-integrated circuit interface (I ² C)	38
3.29	Universal synchronous/asynchronous receiver transmitter (USART)	39
3.30	Low-power universal asynchronous receiver transmitter (LPUART)	40
3.31	Serial peripheral interface (SPI)	40
3.32	Serial audio interfaces (SAI)	41
3.32.1	SAI peripheral supports	41
3.33	Controller area network (FDCAN1)	42
3.34	Universal serial bus (USB)	42
3.35	USB Type-C™ / USB Power Delivery controller (UCPD)	42
3.36	Clock recovery system (CRS)	43
3.37	Development support	43
3.37.1	Serial wire JTAG debug port (SWJ-DP)	43
3.37.2	Embedded trace macrocell™	43
4	Pinouts and pin description	44
4.1	UFQFPN32 pinout description	44

4.2	LQFP32 pinout description	44
4.3	UFQFPN48 pinout description	45
4.4	LQFP48 pinout description	45
4.5	WLCSP49 ballout description	46
4.6	LQFP64 pinout description	46
4.7	LQFP80 pinout description	47
4.8	UFBGA64 ballout description	48
4.9	LQFP100 pinout description	49
4.10	Pin definition	50
4.11	Alternate functions	61
5	Electrical characteristics	67
5.1	Parameter conditions	67
5.1.1	Minimum and maximum values	67
5.1.2	Typical values	67
5.1.3	Typical curves	67
5.1.4	Loading capacitor	67
5.1.5	Pin input voltage	67
5.1.6	Power supply scheme	68
5.1.7	Current consumption measurement	69
5.2	Absolute maximum ratings	69
5.3	Operating conditions	71
5.3.1	General operating conditions	71
5.3.2	Operating conditions at power-up / power-down	72
5.3.3	Embedded reset and power control block characteristics	72
5.3.4	Embedded voltage reference	74
5.3.5	Supply current characteristics	75
5.3.6	Wakeup time from low-power modes and voltage scaling transition times	101
5.3.7	External clock source characteristics	102
5.3.8	Internal clock source characteristics	107
5.3.9	PLL characteristics	110
5.3.10	Flash memory characteristics	111
5.3.11	EMC characteristics	112
5.3.12	Electrical sensitivity characteristics	113
5.3.13	I/O current injection characteristics	114

5.3.14	I/O port characteristics	115
5.3.15	NRST pin characteristics	120
5.3.16	Extended interrupt and event controller input (EXTI) characteristics	121
5.3.17	Analog switches booster	121
5.3.18	Analog-to-digital converter characteristics	122
5.3.19	Digital-to-Analog converter characteristics	137
5.3.20	Voltage reference buffer characteristics	144
5.3.21	Comparator characteristics	147
5.3.22	Operational amplifiers characteristics	148
5.3.23	Temperature sensor characteristics	152
5.3.24	V _{BAT} monitoring characteristics	152
5.3.25	Timer characteristics	153
5.3.26	Communication interfaces characteristics	154
5.3.27	UCPD characteristics	164
6	Package information	165
6.1	UFQFPN32 package information	165
6.2	LQFP32 package information	168
6.3	UFQFPN48 package information	171
6.4	LQFP48 package information	174
6.5	WLCSP49 package information	178
6.6	LQFP64 package information	182
6.7	LQFP80 package information	185
6.8	UFBGA64 package information	188
6.9	LQFP100 package information	191
6.10	Thermal characteristics	194
6.10.1	Reference document	195
6.10.2	Selecting the product temperature range	196
7	Ordering information	198
8	Revision history	199

List of tables

Table 1.	Device summary	2
Table 2.	STM32G431x6/x8/xB features and peripheral counts	14
Table 3.	STM32G431x6/x8/xB peripherals interconnect matrix	25
Table 4.	DMA implementation	27
Table 5.	Temperature sensor calibration values.	30
Table 6.	Internal voltage reference calibration values	30
Table 7.	Timer feature comparison.	32
Table 8.	I2C implementation.	38
Table 9.	USART/UART/LPUART features	39
Table 10.	SAI implementation for the features implementation	41
Table 11.	Legend/abbreviations used in the pinout table	50
Table 12.	STM32G431x6/x8/xB pin definition	51
Table 13.	Alternate function	61
Table 14.	Voltage characteristics	69
Table 15.	Current characteristics	70
Table 16.	Thermal characteristics.	70
Table 17.	General operating conditions	71
Table 18.	Operating conditions at power-up / power-down	72
Table 19.	Embedded reset and power control block characteristics.	72
Table 20.	Embedded internal voltage reference.	74
Table 21.	Current consumption in Run and Low-power run modes, code with data processing running from Flash in single Bank, ART enable (Cache ON Prefetch OFF) . .	76
Table 22.	Current consumption in Run and Low-power run modes, code with data processing running from Flash in single Bank, ART disable	78
Table 23.	Current consumption in Run and Low-power run modes, code with data processing running from SRAM1	80
Table 24.	Typical current consumption in Run and Low-power run modes, with different codes running from Flash, ART enable (Cache ON Prefetch OFF)	82
Table 25.	Typical current consumption in Run and Low-power run modes, with different codes running from Flash, ART disable	83
Table 26.	Typical current consumption in Run and Low-power run modes, with different codes running from SRAM1	84
Table 27.	Typical current consumption in Run and Low-power run modes, with different codes running from SRAM2	85
Table 28.	Typical current consumption in Run and Low-power run modes, with different codes running from CCM	86
Table 29.	Current consumption in Sleep and Low-power sleep mode Flash ON	87
Table 30.	Current consumption in low-power sleep modes, Flash in power-down.	88
Table 31.	Current consumption in Stop 1 mode	89
Table 32.	Current consumption in Stop 0 mode	90
Table 33.	Current consumption in Standby mode	90
Table 34.	Current consumption in Shutdown mode	93
Table 35.	Current consumption in VBAT mode	95
Table 36.	Peripheral current consumption	97
Table 37.	Low-power mode wakeup timings	101
Table 38.	Regulator modes transition times	102
Table 39.	Wakeup time using USART/LPUART.	102
Table 40.	High-speed external user clock characteristics.	102

Table 41.	Low-speed external user clock characteristics	103
Table 42.	HSE oscillator characteristics	104
Table 43.	LSE oscillator characteristics ($f_{LSE} = 32.768$ kHz)	106
Table 44.	HSI16 oscillator characteristics	107
Table 45.	HSI48 oscillator characteristics	108
Table 46.	LSI oscillator characteristics	109
Table 47.	PLL characteristics	110
Table 48.	Flash memory characteristics	111
Table 49.	Flash memory endurance and data retention	111
Table 50.	EMS characteristics	112
Table 51.	EMI characteristics	113
Table 52.	ESD absolute maximum ratings	113
Table 53.	Electrical sensitivities	114
Table 54.	I/O current injection susceptibility	114
Table 55.	I/O static characteristics	115
Table 56.	Output voltage characteristics	117
Table 57.	I/O (except FT_c) AC characteristics	118
Table 58.	I/O FT_c AC characteristics	119
Table 59.	NRST pin characteristics	120
Table 60.	EXTI input characteristics	121
Table 61.	Analog switches booster characteristics	121
Table 62.	ADC characteristics	122
Table 63.	Maximum ADC RAIN	125
Table 64.	ADC accuracy - limited test conditions 1	127
Table 65.	ADC accuracy - limited test conditions 2	129
Table 66.	ADC accuracy - limited test conditions 3	131
Table 67.	ADC accuracy (Multiple ADCs operation) - limited test conditions 1	133
Table 68.	ADC accuracy (Multiple ADCs operation) - limited test conditions 2	134
Table 69.	ADC accuracy (Multiple ADCs operation) - limited test conditions 3	135
Table 70.	DAC 1MSPS characteristics	137
Table 71.	DAC 1MSPS accuracy	140
Table 72.	DAC 15MSPS characteristics	141
Table 73.	DAC 15MSPS accuracy	143
Table 74.	VREFBUF characteristics	144
Table 75.	COMP characteristics	147
Table 76.	OPAMP characteristics	148
Table 77.	TS characteristics	152
Table 78.	V _{BAT} monitoring characteristics	152
Table 79.	V _{BAT} charging characteristics	152
Table 80.	TIMx characteristics	153
Table 81.	IWDG min/max timeout period at 32 kHz (LSI)	154
Table 82.	WWDG min/max timeout value at 170 MHz (PCLK)	154
Table 83.	Minimum I2CCLK frequency in all I2C modes	155
Table 84.	I2C analog filter characteristics	155
Table 85.	SPI characteristics	156
Table 86.	I2S characteristics	159
Table 87.	SAI characteristics	161
Table 88.	USB electrical characteristics	163
Table 89.	USART electrical characteristics	163
Table 90.	UCPD characteristics	164
Table 91.	UFQFPN32 - mechanical data	166
Table 92.	LQFP32 - mechanical data	169

Table 93.	UFQFPN48 - mechanical data	172
Table 94.	LQFP48 - mechanical data	175
Table 95.	WLCSP49 - mechanical data	179
Table 96.	WLCSP49 recommended PCB design rules	180
Table 97.	LQFP64 - mechanical data	182
Table 98.	LQFP80 - mechanical data	185
Table 99.	UFBGA64 - mechanical data	188
Table 100.	UFBGA64 recommended PCB design rules (0.5 mm pitch BGA)	189
Table 101.	LQPF100 - mechanical data	191
Table 102.	Package thermal characteristics	194
Table 103.	Ordering information scheme	198
Table 104.	Document revision history	199

List of figures

Figure 1.	STM32G431x6/x8/xB block diagram	16
Figure 2.	Multi-AHB bus matrix	19
Figure 3.	Voltage reference buffer	31
Figure 4.	Infrared transmitter	37
Figure 5.	STM32G431x6/x8/xB UFQFPN32 pinout	44
Figure 6.	STM32G431x6/x8/xB LQFP32 pinout	44
Figure 7.	STM32G431x6/x8/xB UFQFPN48 pinout	45
Figure 8.	STM32G431x6/x8/xB LQFP48 pinout	45
Figure 9.	STM32G431x6/x8/xB WLCSP49 ballout	46
Figure 10.	STM32G431x6/x8/xB LQFP64 pinout	46
Figure 11.	STM32G431x6/x8/xB LQFP80 pinout	47
Figure 12.	STM32G431x6/x8/xB UFBGA64 ballout	48
Figure 13.	STM32G431x6/x8/xB LQFP100 pinout	49
Figure 14.	Pin loading conditions	67
Figure 15.	Pin input voltage	67
Figure 16.	Power supply scheme	68
Figure 17.	Current consumption measurement	69
Figure 18.	VREFINT versus temperature	75
Figure 19.	High-speed external clock source AC timing diagram	103
Figure 20.	Low-speed external clock source AC timing diagram	103
Figure 21.	Typical application with an 8 MHz crystal	105
Figure 22.	Typical application with a 32.768 kHz crystal	106
Figure 23.	HSI16 frequency versus temperature	108
Figure 24.	HSI48 frequency versus temperature	109
Figure 25.	I/O input characteristics	116
Figure 26.	I/O AC characteristics definition ⁽¹⁾	120
Figure 27.	Recommended NRST pin protection	121
Figure 28.	ADC accuracy characteristics	136
Figure 29.	Typical connection diagram using the ADC	136
Figure 30.	12-bit buffered / non-buffered DAC	139
Figure 31.	VREFOUT_TEMP in case VRS = 00	145
Figure 32.	VREFOUT_TEMP in case VRS = 01	146
Figure 33.	VREFOUT_TEMP in case VRS = 10	146
Figure 34.	OPAMP noise density @ 25°C	151
Figure 35.	SPI timing diagram - slave mode and CPHA = 0	157
Figure 36.	SPI timing diagram - slave mode and CPHA = 1	158
Figure 37.	SPI timing diagram - master mode	158
Figure 38.	SAI master timing waveforms	162
Figure 39.	SAI slave timing waveforms	162
Figure 40.	UFQFPN32 - outline	165
Figure 41.	UFQFPN32 - recommended footprint	166
Figure 42.	UFQFPN32 top view example	167
Figure 43.	LQFP32 - outline	168
Figure 44.	LQFP32 - recommended footprint	169
Figure 45.	LQFP32 top view example	170
Figure 46.	UFQFPN48 - outline	171
Figure 47.	UFQFPN48 - recommended footprint	172
Figure 48.	UFQFPN48 top view example	173